BrytEnergy

D55

DATA SOVEREIGNTY ASSESSMENT

Contract Note Template Management

Service Interaction & Data Sovereignty (GDPR) Assessment

Confidential - Bryt Energy & D55

D55

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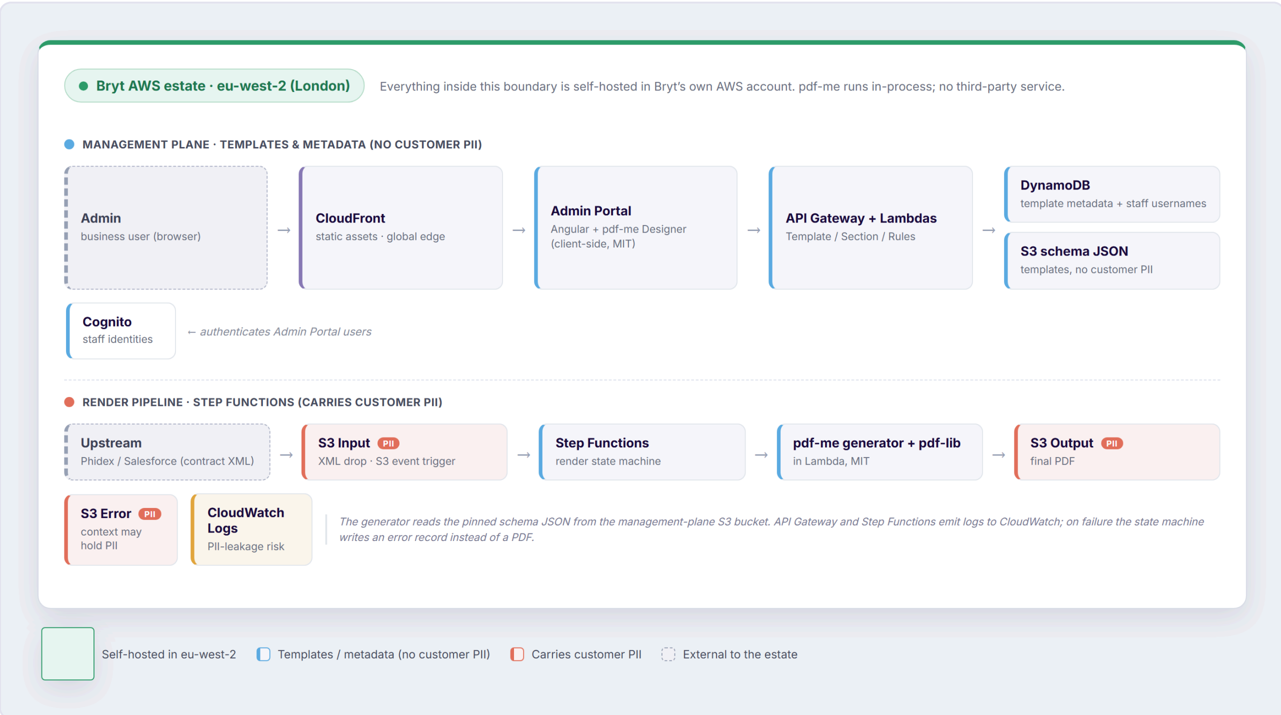
Overview

This assessment covers the Bryt Energy Contract Note Rework: the template management system and serverless render pipeline built with **pdf-me**. It sets out the main service interactions and evaluates whether the design raises any data-sovereignty or GDPR concerns, given that pdf-me is open source and deployed entirely within Bryt's own AWS estate in the **eu-west-2 (London)** region.

The headline: the entire solution is **self hosted, safe and defensible**. The residual risk is not pdf-me or the region; it is the operational plumbing around the two buckets that carry customer PII, tracked as caveats at the end of this document.

Service interaction

The diagram below shows the main pieces of the solution and, critically, **where customer personal data actually flows**. Nodes tinted red carry customer PII; blue nodes carry templates and metadata only; everything inside the green boundary is self-hosted in Bryt's own AWS account in eu-west-2.



High-level service interaction, showing the eu-west-2 boundary and the paths that carry customer PII.

The two flows are deliberately separated. The **management plane** (top) is where business users author templates; it handles layout and metadata, not customer data. The **render pipeline** (bottom) is where contract data flows: an XML drop triggers the Step Functions state machine, which renders sections with the pdf-me generator and stitches the final PDF. Only the input, output and error buckets in that lower lane hold customer PII.

Key facts

pdf-me is a library, not a service.

It is MIT-licensed and open source. The Designer runs client-side in the admin's browser; the generator runs in-process inside a Lambda. No contract data is sent to pdfme.com or any external endpoint.

No new processor or sub-processor.

Because pdf-me executes inside Bryt's own compute, it introduces no third party to the data flow; nothing to add to a GDPR processor register or a data-processing agreement.

Everything runs in eu-west-2.

S3 (input, output, error), DynamoDB, the Lambdas and Step Functions are all in the London region, so customer PII is stored and processed in the UK.

The lawful basis for location holds.

Under UK GDPR this is domestic processing. For EU data subjects, the UK holds an EU adequacy decision, so transfers are lawful with no additional safeguards required.

The PII footprint is narrow.

Customer PII lives only in the XML input, the PDF output and (potentially) the error bucket. The template, schema and metadata stores carry layout, not customer data.

Overall position

The self-hosted, MIT-licensed, eu-west-2 argument is sound. pdf-me does not weaken data sovereignty because it never receives data outside the estate, and the region choice keeps all processing in the UK under an adequate legal basis. What remains is a short list of operational controls that an auditor or DPO would expect to see evidenced; none are blockers, but they are where a "self-hosted so we're fine" assumption typically springs a leak.

Caveats to close out

Each item below carries a risk rating and a note on the evidence that closes it. Four have been reviewed and verified; three remain in progress, as flagged in the Verification column.

#	CAVEAT	RISK	STATUS	EVIDENCE THAT CLOSES IT	VERIFICATION
1	CloudWatch log leakage	High	Investigating	Confirm Lambdas do not log parsed contract data; review the error bucket's open-ended `context` field for PII; set explicit log-retention periods.	Investigating
2	Font loading stays in-estate	Medium	Closed	Verify pdf-me fonts (e.g. NotoSans) are bundled into the Lambda / front-end bundle, not fetched from a Google Fonts CDN at render time.	Checked OK
3	CloudFront edge caching	Low	Closed	Confirm no API / PII responses are cached at the global edge; only static, non-personal assets are distributed.	Checked OK
4	Region pinning of all stores	Medium	Closed	Verify no DynamoDB global tables, no cross-region S3 replication, and that any backup / DR target is in-region (or another adequate jurisdiction).	Checked OK
5	Encryption at rest and in transit	Low	Investigating	Confirm S3 and DynamoDB encryption at rest (KMS) and TLS on API Gateway; record for the audit trail.	S3 OK Dynamo Investigating

#	CAVEAT	RISK	STATUS	EVIDENCE THAT CLOSES IT	VERIFICATION
6	Retention & right to erasure	High	Needs discussion	Define lifecycle / retention policies on the output bucket and a deletion path, to satisfy storage-limitation and right-to-erasure obligations.	Needs Discussion
7	Preview with real customer data	Medium	Closed	Ensure the Designer previews templates with sample / synthetic data only, so live PII never lands in an admin's browser.	Checked OK

Notes

- **pdf-me licensing:** MIT, per the project repository (github.com/pdfme/pdfme).
- **Region:** all stateful services in eu-west-2 (London); adequacy decision applies for EU data subjects.